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SURFACE RECONSTRUCTION AND SURFACE EXPLOSION PHENOMENA IN THE NICKEL (110)/HYDROGEN SYSTEM

K. CHRISTMANN *, F. CHEHAB **, V. PENKA and G. ERTL

Institut fur Physikalische Chemie, Universität München, Sophienstrasse 11, D-8000 München 2, Fed. Rep. of Germany

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Combined Video–LEED, work function ($\Delta\phi$), thermal desorption (TD) and UV photoelectron spectroscopy (UPS) measurements revealed a variety of hydrogen assisted phase transformations on a Ni(110) surface between 120 and 250 K. Among others, a 2×1-2H lattice gas phase (at a coverage $\theta_H = 1$ ML) undergoes a first-order transition to a reconstructed 1×2 phase with $\theta_H = 1.5$ ML at saturation. In UPS this phase produces a strong extra emission near -1.3 eV below the Fermi energy. All low temperature phases are however only metastable and suffer an irreversible transition to a merely one-dimensionally ordered "streak" phase (which is likewise reconstructed) as the surface is heated to beyond ~ 200 K. The transition $1\times2 \rightarrow$ "streak" occurs very rapidly in a narrow temperature range and is characterised by a sudden break-down of the LEED intensity of the extra spots and a steplike decrease of the work function. Also, the photoemission features 1.3 eV below E_F disappear completely. Moreover, the transition is accompanied by an explosive evolution of hydrogen which desorbs in a sharp α-state thereby suggesting the decomposition of a surface compound. Structure models and mechanisms are presented and discussed in order to rationalise the experimental findings.

1. Introduction

Two-dimensional phase transitions in adsorbed layers offer the possibility to determine lateral interaction energies between adatoms or to test theoretical models of ordering phenomena in two dimensions [1]. There are various categories of two-dimensional phase transitions; with regard to the Ni(110)/H system two of them are of particular interest:
(1) The transition occurs within the layer of adsorbed atoms only, and the substrate atoms do not change their positions upon adsorption. They rather provide a rigid lattice of sites which can be occupied or unoccupied. Usually, a so-called "lattice gas model" is appropriate to describe the configuration of the

* Institut für Physikalische Chemie, Freie Universität Berlin, D-1000 Berlin 33, Germany.
** Institut für Physikalische Chemie, Universität Hamburg, D-2000 Hamburg 13, Fed. Rep. of Germany.

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adatoms as a function of the temperature T and the surface coverage θ, i.e., the chemical potential μ of the adsorbate.
(2) The presence of the adatoms on the surface does affect the position of the topmost substrate atoms. At a certain critical coverage θ_c or temperature T they undergo a phase transition. This phenomenon is well-known as adsorbate-induced reconstruction and has been dealt with in the literature many times [2]. Apparently strongly interacting adatoms are required which perturb the electronic structure of the metal sufficiently to make its surface atoms move to new equilibrium positions.

The Ni(110)/H system is unique in that it offers a good example for both types of phase transitions. As reported elsewhere [3], lattice gas phases form in the submonolayer coverage range θ_H as long as the surface temperature is kept below 130 K. With these phases the Ni surface remains essentially undistorted. For appreciable θ_H, however, viz. for $\theta_H > 1$ ML, a H induced surface reconstruction to a 1×2 phase takes place which is completed at $\theta_H = 1.5$ ML. Another peculiar phase transformation occurs if the sample is heated to near 200 K: No matter which surface phase exists initially, a reconstructed streak phase with only one-dimensional periodicity forms – this represents the only stable surface phase in the presence of hydrogen and at temperatures above 230 K. Superimposed on this transition is the rapid decomposition of the 1×2 phase at 220 K which resembles a surface melting process with the exception that it is essentially irreversible, that is, cooling to below 220 K does not restore the 1×2 phase anymore as long as the streak phase is present.

We have performed Video–LEED, $\Delta\phi$, TDS and UPS measurements in order to elucidate the nature of these phases and their transitions. A more extensive report of our findings is given elsewhere [4].

2. Experimental

The experiments were performed in a standard stainless steel vacuum chamber equipped with the usual facilities to prepare and characterise a clean metal surface, viz., Video–LEED, Auger electron spectroscopy (AES), thermal desorption spectroscopy (TDS) utilising a quadrupole mass spectrometer and a Kelvin capacitor to record work function changes ($\Delta\phi$). Angular resolved UPS measurements using He I and II radiation were carried out in a second chamber with similar surface analysis techniques. A liquid N_2 cooling manipulator allowed experiments at 120 K. Further details of the experimental set-up can be taken from previous papers [4,5]. The sample cleaning turned out to be tedious. It was accomplished by prolonged argon ion sputtering and subsequent flashes in 10^{-6} Torr oxygen atmosphere. Particularly the submonolayer lattice gas phases of hydrogen could only be observed if the surface was carefully cleaned. The order of the surface phases could be conveniently

monitored by means of the Video–LEED method [6] whereby an electronic window was generated and set around the LEED spots of interest and the intensity was integrated and analysed using a LSI 11 computer. The advantage of the fast Video–LEED method is the substantial reduction of electron beam effects which is especially helpful when studying the sensitive H adsorbed phases.

3. Results

3.1. LEED

The clean Ni(110) sample was exposed to increasing amounts of hydrogen at 120 K. As described previously [3,7] various lattice gas phases form in the submonolayer range, the final phase being a 2×1-2H with $\theta_H = 1.0$ ML reached after 0.7 L exposure. An exposure of 2.5 L is sufficient to complete the $2 \times 1 \rightarrow 1 \times 2$ phase transition (which is a first-order transition) and to produce a bright LEED pattern with sharp and intense "extra" spots of a 1×2 structure. This is schematically shown in fig. 1a.

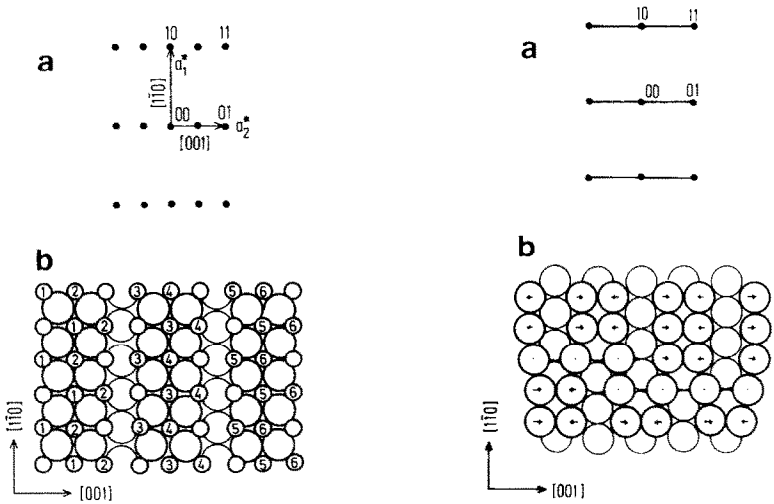


Fig. 1. (a) Schematic LEED pattern of the low-temperature 1×2 phase. (b) Corresponding structure model with the three-fold and four-fold sites indicated by arrows. The numbers 1–6 indicate the zig-zag rows of the H atoms in the 2×1 phase.

Fig. 2. (a) Schematic LEED pattern of the high-temperature "streak" phase. (b) Corresponding structure model with the displacements of the Ni atoms in [001] direction indicated by arrows. Location of H atoms not shown.

Any hydrogen exposure at temperatures above ~ 200 K however leads to the formation of a different phase with sharp and intense streaks in [001] direction as shown in fig. 2a. Another way to form this "streak" structure is to heat any low temperature H phase to beyond ~ 220 K: Always the streak structure appears as the final stable phase. The Video–LEED technique enabled us to monitor the intensity of the H induced "extra" features as a function of both the coverage and the temperature. Here, we concentrate on the temperature dependence whilst the θ_H dependence is dealt with in a previous publication [3].

Fig. 3 shows the T dependence of the intensity in the $(0, \frac{1}{2})$ position of the reciprocal space. After setting the appropriate electronic window the temperature of the sample was slowly raised and the intensity integrated. The two curves shown in fig. 3 stand for two different initial hydrogen coverages: The dashed line refers to $\theta_{H,i} = 0.5$; the full line to $\theta_{H,i} = 1.5$ ML, i.e., to the fully developed 1×2 phase. In case of the $\theta_H = 0.5$ curve the (streak) intensity is zero up to 150 K but increases continuously as the temperature raises, passes through a maximum around 320 K and thereafter falls again owing to thermal desorption of the adsorbed hydrogen. A different situation is observed with the 1×2 phase, i.e., $\theta_{H,i} = 1.5$. Here there is a pronounced intensity (the 1×2 "extra" intensity) already at 120 K which remains almost constant up to ~ 180 K before it decays somewhat and finally drops sharply when the temperature reaches a value of 220 K. It does not go back to zero but rather reaches the intensity of the streak phase suggesting that the 1×2 phase transforms directly to the streak phase. There is however some indication that the 1×2 phase does not directly transform to the streak phase; from measurements of the integral

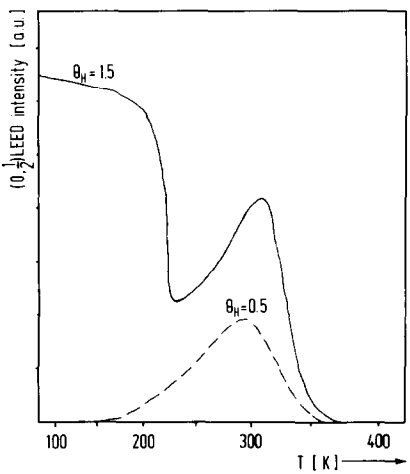


Fig. 3. Temperature dependence of the LEED intensity in $(0, \frac{1}{2})$ position of the reciprocal lattice for two different initial coverages: (– – –) $\theta_{H,i} = 0.5$ ML; (———) $\theta_{H,i} = 1.5$ ML (1×2 phase).

order beams of the Ni substrate it rather appears as if momentarily an unreconstructed 1×1 phase is passed which rapidly converts to the streak phase which represents the equilibrium structure under these conditions. Once the streak phase is formed there is no way to regenerate the sharp 1×2 phase or the lattice gas phases by simply cooling the streak surface in hydrogen to below 200 K. Rather this phase persists and even increases slightly in intensity. In order to produce the other phases all the hydrogen has to be thermally desorbed first, then the sample has to be exposed to H_2 at 120 K. Thus we conclude that the formation of the streak structure is an irreversible process. There is a second irreversible surface reaction, namely, the formation of the 1×2 structure. Even at a high ambient pressure of hydrogen this phase cannot be formed from the streak phase: The regular order of the 2×1-2H lattice gas phase is required to form ther 1×2 phase, for reasons explained in the discussion, section 4.

3.2. TDS measurements

As compared to the other low-index planes of Ni the TD behaviour of the Ni(110) surface appears to be quite complex. A series of TD curves displayed in fig. 4 demonstrates this: Three distinct desorption states are observed at full coverage, namely, two β-states (β_1 and β_2) and a sharp α-state at somewhat lower temperatures. All these states reflect a competition between thermal desorption processes and (simultaneously occurring) surface reconstruction phenomena which are thermally activated. As described elsewhere in more detail [4] there is a partial formation of the streak phase *during* the application of the temperature program which leads to a surface which consists partly of not yet reconstructed areas (from these desorption occurs in the β_1-state) and partly of already reconstructed (streaky) areas (from which H_2 desorbs in the β_2-state). The α-state can only be observed when the 1×2 phase is present; it is obviously associated with the decomposition of this phase because it appears right at 220 K where the 1×2 LEED intensity disappears. Apart from the very narrow half-width of this state it is noteworthy that its desorption kinetics obeys a fractional order rate law. This is indicative of a decomposition reaction of a surface phase in which islands dissolve at their perimeters. A more detailed description and consideration of these TDS phenomena can be found elsewhere [4]; we only add here that about 30% of the adsorbed hydrogen desorbs in the α-state. Integration of the TD peak areas also confirms the assumption that the coverage associated with the 2×1-2H phase is 1.0 ML, and that associated with the complete 1×2 phase is 1.5 ML, in agreement with previous [7] and recent [8] coverage determinations. From our TDS measurements also kinetic data are available; we simply note here that the initial sticking probability at 120 K is 0.87 ($\pm 20\%$), in good agreement with previously reported values [9].

3.3. $\Delta\phi$ measurements

The LEED and TDS observations are supported and further confirmed by our work function measurements. Again, only a brief survey of the $\Delta\phi$ results is given here. The work function increases upon adsorption of hydrogen, in agreement with previous studies [10,11], by a total amount of 510 mV at 120 K at saturation, that is to say, with the 1×2 phase fully developed. The $\Delta\phi$ of 510 mV includes *two* contributions, namely, the work function produced by the H dipoles, and the work function caused by the surface reconstruction. If the 1×2 surface is heated up from 120 K there is first only little effect on $\Delta\phi$; near 220 K, however, we find a steplike decrease by about 80 mV, exactly at the temperature where the 1×2 LEED intensity broke down and the sharp

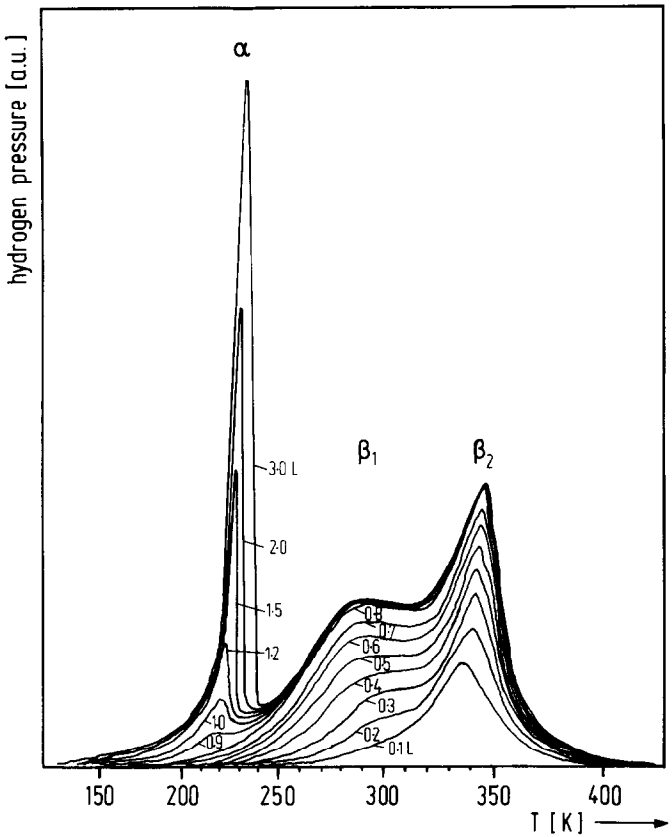


Fig. 4. Series of thermal desorption spectra obtained after adsorption of hydrogen at 120 K. The heating rate was ~ 9 K/s. The various desorption states are indicated. Note the extreme sharpness and peculiar kinetics of the α-state.

α-TD state appeared. Again, there is a pronounced irreversibility with regard to the formation of the 1×2 phase: Heating to 300 K in a hydrogen atmosphere and cooling down to below 220 K does not lead to the initial 510 mV but only to ~ 420 mV work function change. This indicates that the 1×2 phase cannot be generated from the streak phase (which is formed by heating to 300 K) and the 80–90 mV associated with the 1×2 phase are simply lacking. A more extensive report on the work function data is given elsewhere [4].

3.4. UV photoemission measurements

Since these results will be published elsewhere in greater detail [12] we only present a brief summary here. The essence of the photoemission measurements can be taken from figs. 5a–5d. It shows (in normal emission) He I spectra of a Ni(110) surface covered with increasing amounts of hydrogen. To demonstrate the effects more clearly, difference spectra are shown where the energy distribution curves of the clean surface has been subtracted from curves b–d. As fig. 5b reveals there is hardly an effect of the H adsorption in the 2×1

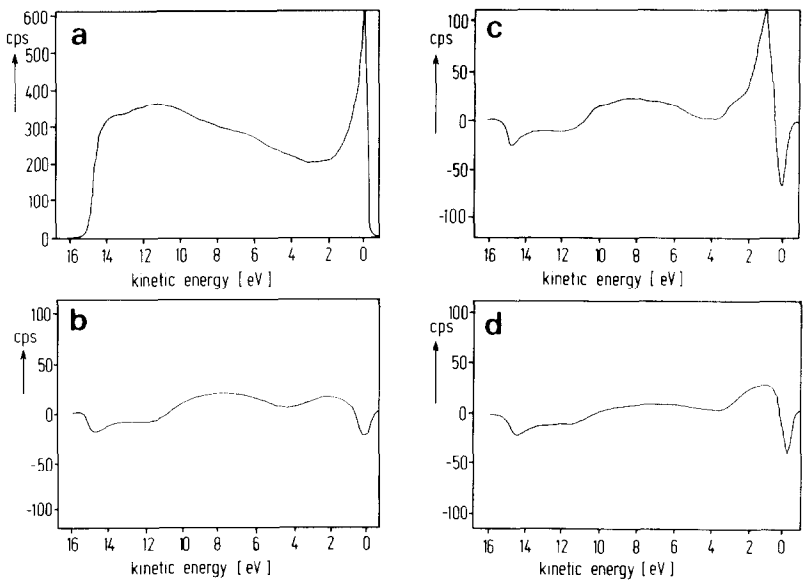


Fig. 5. Sequence of UV photoemission energy distribution curves (normal emission) from a Ni(110) surface: (a) clean surface; (b) covered with 1 ML of H (2×1 phase); difference spectrum; (c) covered with 1.5 ML H at 120 K (1×2 phase fully developed; difference spectrum; (d) covered with 1.5 ML H at 120 K, heated to 250 K to induce the phase transformation to the streak structure and cooled down; θ_H between 1.0 and 1.5 ML; difference spectrum.

phase ($\theta_H = 1$) on the UPS curves, at least not in normal emission. However, as soon as the 1×2 reconstructed phase starts to form (fig. 5c) a new state in the d-band region of the Ni approximately 1.3 eV below the Fermi energy develops. That this state is associated with the 1×2 phase can immediately be seen if the surface is heated to beyond 220 K (where, as we know, the 1×2 phase decomposes): The new state disappears completely and is not noticeably restored if H is post-adsorbed at lower temperatures. The streak phase, on the other hand, gives at most rise to a slight shoulder around 1–1.5 eV below E_F but no distinct state as the 1×2 phase. In order to monitor the energy–parallel momentum ($E(k_{\|})$) dependence we have varied the polar angle of the incident light for different (constant) azimuth angles Φ, namely, $\Phi = 0°$ (corresponding to the ΓK direction = $[1\bar{1}0]$), and $\Phi = 90°$ (corresponding to the ΓX direction = [001]). For the 2×1 phase at $\theta_H = 1.0$ (which did not produce significant extra features in normal emission) a vague state becomes apparent as the parallel momentum $k_{\|}$ is increased. Interestingly, this state does not show a dispersion in ΓX direction but exhibits a noticeable dispersion in ΓX direction, i.e., parallel to the densely packed rows of the Ni atoms. The state -1.3 eV below E_F associated with the 1×2 phase does not exhibit any noticeable dispersion, neither in ΓX nor in ΓK direction. Furthermore, the splitting of the upper Ni d-band in ΓK and Γx direction is strongly suppressed when the 1×2 phase forms, thus indicating that this latter phase perturbs the electronic structure of the Ni surface strongly.

4. Discussion

Insight into the above phase transformations requires a knowledge of the *structure* of the phases involved. Despite the vast literature on Ni(110)/H [3–13], precise information about the actual geometry of these phases is not available. Only recently, He diffraction [7] and LEED [3] experiments could elucidate the nature of some of the phases. As suggested by Tucker [14] a pairing of (110) Ni row atoms in [001] direction is responsible for the 1×2 phase. A local 1×1 H structure involving also the second layer atoms of the Ni surface follows from LEED [5] and the coverage of $\theta_H = 1.5$ can only be reconciled if the H atoms are placed in threefold and fourfold adsorption sites as schematically depicted in fig. 1b.

Even less structural information exists for the "streak" phase although this is known since the early days of LEED [13]. Certainly this is also a reconstructed phase: The weak scattering power of adsorbed H atoms for low-energy electrons cannot produce streaks of such high intensity. The LEED pattern associated with the streak phase suggests perfect periodicity in $[1\bar{1}0]$ direction and random periodicity in the perpendicular [001] direction. A persuasive structure model is given in fig. 2b: As compared to the 1×2 phase

with its very regular coupling of Ni pairs in [001] direction this periodicity is entirely lost in the streak phase. Instead, such pairs may well be formed *locally*, with no long-range correlation in [001] direction. If it is assumed that always two adjacent pairs form a fourfold hollow site with an adsorbed H atom in the centre, a situation similar to that of the Ni(100)/H system [15] arises. Overall, between 0.25 and 0.50 as many fourfold sites as there are displaced Ni atoms are generated, if we assume statistically displaced Ni atoms in the streak phase. In the 1×2 phase with its maximum number of fourfold sites (namely, 0.5 times the number of displaced Ni atoms) the hydrogen coverage is 1.5. Thus we expect a somewhat reduced H coverage for the streak phase, i.e., $1.0 < \theta_H < 1.5$. Experimentally this is indeed observed as the desorption of the α-state demonstrates. The remaining hydrogen desorbs in the β_2-state, and its desorption temperature is very close to that found with the Ni(100)/H system [16].

The observed structural transition from the 1×2 phase to the final streak phase is not a reversible (i.e., equilibrium) transition. Rather it is determined by kinetic factors, e.g., nucleation phenomena. As will be shown below we may consider the 1×2 phase as a surface hydride compound. If its decomposition (as indicated by the explosive evolution of hydrogen in the α-state) would only start at few points to which the H atoms have to migrate a zero-order desorption kinetics would arise. If, on the other hand, this decomposition process starts at a critical temperature (like a melting process) more or less randomly, the reaction order would be between 0 and 1 as actually observed. As a consequence necessarily *isolated* fourfold hollow sites will be left which decompose at higher temperatures: The streak phase is formed which can only be destroyed by thermal desorption of H in the β_2-state. Now we consider the reverse process of adsorption. The pairing of neighbouring Ni atoms is certainly associated with an activation barrier. At low temperatures it cannot be overcome *unless the 2×1-2H phase forms* which leads to a sufficient weakening of the Ni–Ni bonds so that the energy gain due to the additional uptake of hydrogen causes the reconstruction to the 1×2 phase or, in other words, the formation of the surface hydride compound. It is immediately seen that the 2×1 phase is a necessary *precursor* for the formation of the 1×2 structure. Since the 2×1 phase disappears (or at least disorders) around 180 K the 1×2 formation cannot take place at higher temperatures. What happens then at temperatures higher than 220 K is that the activation energy for *local* pair formation can be overcome, and the creation of locally more or less isolated fourfold sites becomes possible. Since, however, this process does not start from an ordered 2×1 phase the pairs formed will necessarily be out of phase: These together with the undisplaced Ni atoms make up the streak phase illustrated in fig. 2b.

A somewhat speculative assumption is the formation of a Ni surface hydride Ni_xH_y associated with the 1×2 phase. Not only can this explain all observed transition phenomena but, more importantly, can be regarded as the *driving*

force for the formation of the 1×2 phase, for simple energetic reasons. Gibb [17] has reviewed the physical chemistry of the primary hydrides; in his article he questioned the existence of *bulk* Ni hydride. The reason was sought from a competition between the Me–H and the Me–Me bond formation: The stability of the latter was assumed to increase with the number of available d-electrons and vice versa. Accordingly, the transition metals of the right-hand side of the periodic table do not form stable metallic hydrides (Pd and Cu are exceptions), whereas the left-hand members (Ti, V, Zr) do. With Ni(110)/H, the 1×2 reconstruction makes adjacent Ni atoms move together and, at the same time, *decouples* these from their neighbours on the other side of the row. Here, the d-electron overlap is substantially perturbed and the lateral metallic bond weakened, a phenomenon referred to as H-induced "decohesion" [18]. A decrease of the strength of the metallic bond will, on the other hand, *improve* the stability of a hydridic bond, and therefore Ni hydride may locally form as a *surface* compound. Unfortunately its structure and stoichiometry cannot be deduced from the present experiments; this will be difficult with any experimental method. There are some important hints though, one being the unusual sharpness of the α-state and its peculiar desorption kinetics (which is reminiscent of a process that has been called a surface "explosion" [19]). A "surface melting" of the hydride compound and thermal decomposition could easily account for these phenomena. The other hint is the occurrence of a well-defined electronic state associated with the 1×2 phase in the d-band region 1.3 eV below E_F. It could either reflect a general rearrangement of the Ni d-electrons in the 1×2 phase or be caused by a hydridic compound which is largely decoupled from the Ni bulk underneath. The strong perturbations of the Ni electronic structure associated with the 1×2 phase could well point into this direction.

In conclusion, we have shown that there is a peculiar irreversible phase transition occurring in the Ni(110)/H system which is not governed by equilibrium thermodynamics but rather may be a consequence of the irreversible decomposition of a surface compound (Ni hydride) which forms in a first-order process at $\theta_H = 1.0$ at 120 K. Structural requirements to form that compound can be made responsible for the irreversibility of the transition at 220 K. We offer a model to rationalise the structure of the "streak" phase formed by exposure of hydrogen to aNi(110) surface above 250 K: Pairs of Ni atoms are statistically formed which are out of phase in [001] direction.

Acknowledgements

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